

# Evaluating Math & Literacy Growth During the Mindsets!® Math Program

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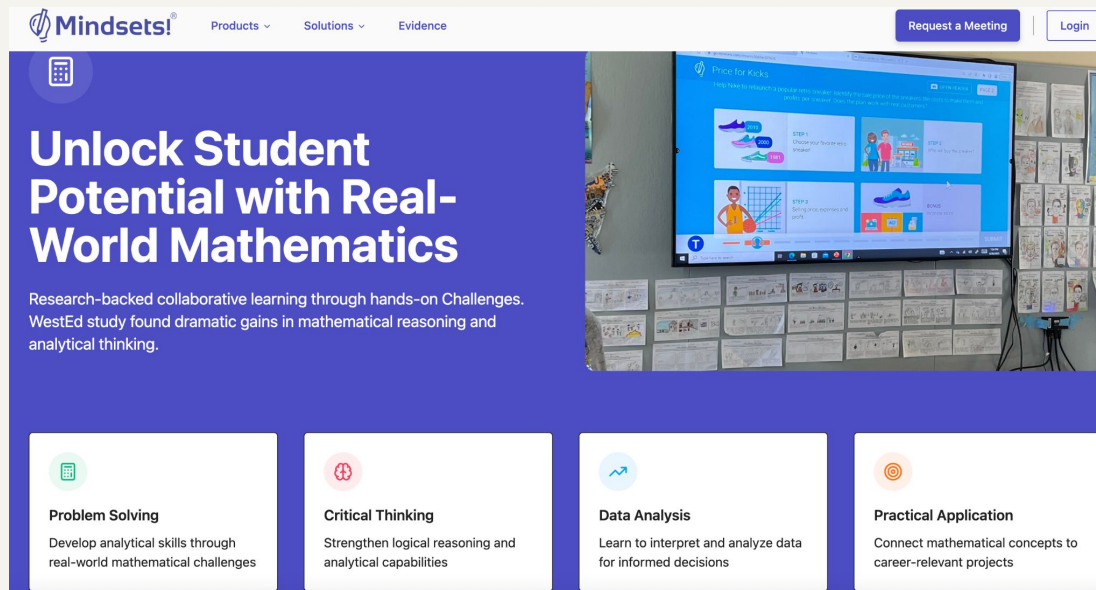
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Capstone Project 2026

# Overview

- Does math predict literacy?
- Does literacy predict math?

# Defining Mindsets!®

- EdTech company
- Project-based learning
- **Math** program with **literacy** supports



# Sample

- Complete attendance records varied by study:
  - Math, Literacy, Other:  $N = 103$
  - Math only:  $N = 107$
  - Literacy only:  $N = 241$
- No assumption of generalization to incomplete attendance records
- 3 waves: 5 weeks in Summer 2024
- Mixed grades 6-8 within student work groups, large urban district

# Data

- Scores: Internal assessment, % correct
- Pre-cleaned data
- Covering “Reason abstractly and quantitatively” (SMP2), “Look for and express regularity in repeated reasoning” (SMP8), literacy, and completion rate/stage attainment measures.

# SMP2 Definition

California Department of Education, *Standards for Mathematical Practice*

## **“Reason abstractly and quantitatively” (SMP2)**

**How many buses are needed for 99 children if each bus seats 44? A student might write:  $99 \div 44$  and find 2 remainder 11. However, the answer is not “2 remainder 11 buses.” Since 11 children are still left without seats, the answer must be 3 buses.**

# SMP8 Definition

**“Look for and express regularity in repeated reasoning.” (SMP8)**

**Nina added several pairs of multiples of 6:**

**$6+12=18$ ,  $18+24=42$ ,  $12+30=42$ ,  $24+36=60$**

**She noticed: "Every time, the sum is also a multiple of 6." Explain to Nina why this will always happen.**

**Any multiple of 6 can be written as 6 times a whole number, so adding two multiples of 6 means adding  $6(a) + 6(b) = 6(a+b)$ , which is still 6 times a whole number and therefore a multiple of 6.**

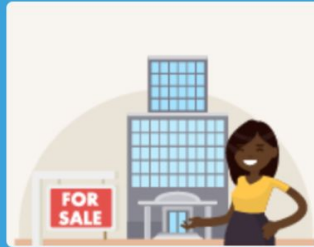
Sources: Standards wording from the California Department of Education. SMP2 example adapted from the Education Development Center; SMP8 task adapted from Illustrative Mathematics.

# Student Interface

Welcome...

11 Questions | 7 Bonus | 4 Polls

On the TV show House Hunters, you sell the most amazing homes in the world. How much could you earn from sales you make on the show?



## STEP 1

Sell your first property.



## STEP 2

What effect does a drop in price have on your commission?



## STEP 3

Decide between a % commission or fixed dollar amount.



## BONUS

You expand internationally, but which country?

Screenshot  
from 7th  
Grade  
Challenge on  
Mindsets  
platform

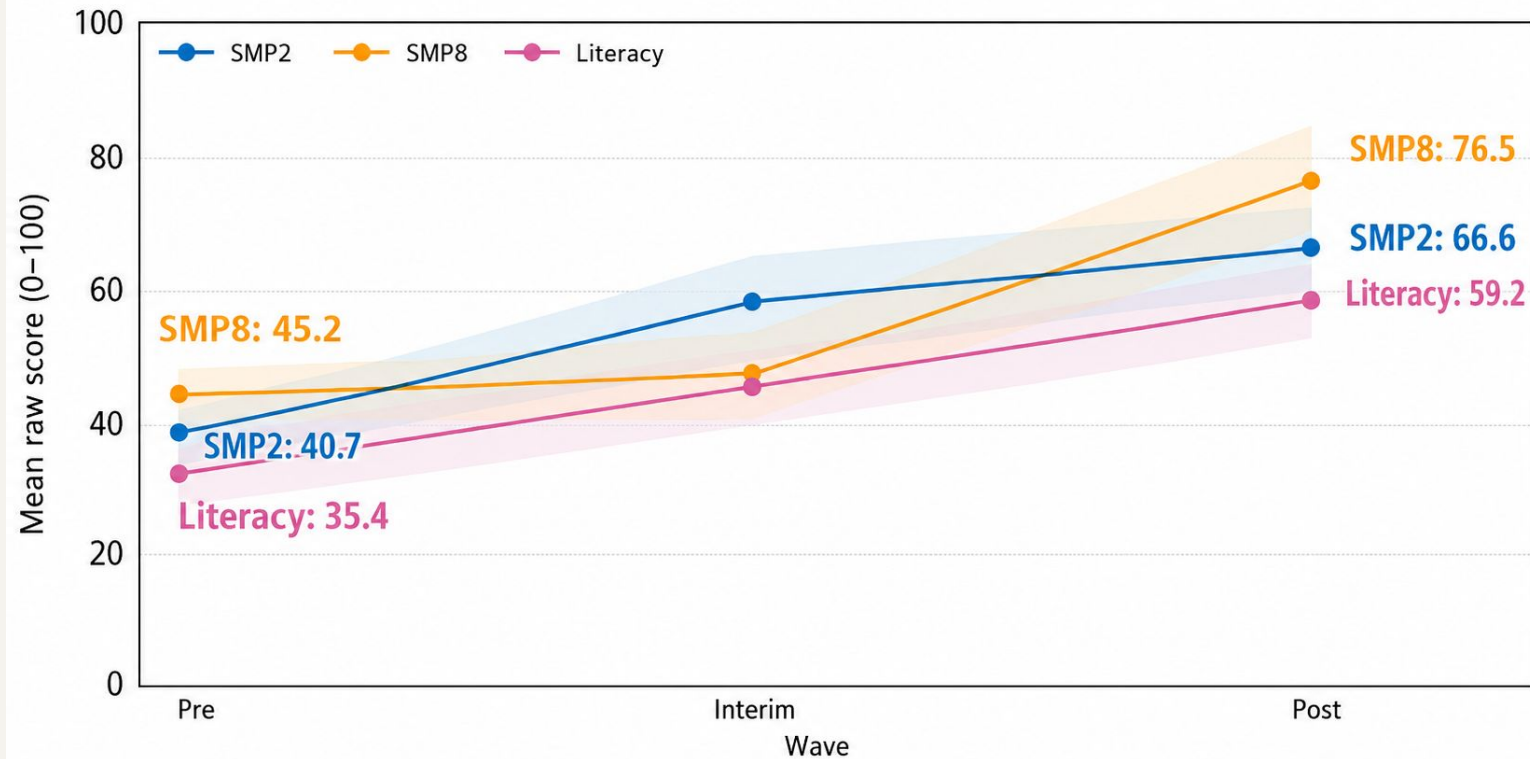




# Research Questions

1. *Math*: Did students demonstrate significant pre-to-post changes in SMP2 and SMP8 performance? Do abstract & quantitative math reasoning (SMP2) support repeated math reasoning (SMP8) more strongly than the reverse?
2. *Literacy*: How did literacy scores change across the program, and did students differ in their growth trajectories?
3. *Math & Literacy*: Do earlier literacy scores predict later math growth? Do earlier math scores predict later literacy growth?
4. *Math, Literacy, Mediator*: Do completion and stage attainment mediate the relationship between growth in academic domains?

# Observed Mean Scores Across Program Waves



**SMP 2:**  
Abstract &  
Quantitative  
Reasoning

**SMP8:**  
Repeated  
Reasoning

**Literacy**

# Research Question 1: Math

- Did students demonstrate significant pre-to-post changes in SMP2 and SMP8 performance during participation in the Mindsets math program?
- Do abstract & quantitative math reasoning (SMP2) predict repeated math reasoning (SMP8)?
- Is the reverse true?
- What is the best order to teach SMP2 and SMP8?

# Research Question 1 Method

## **Paired Samples *t*-test**

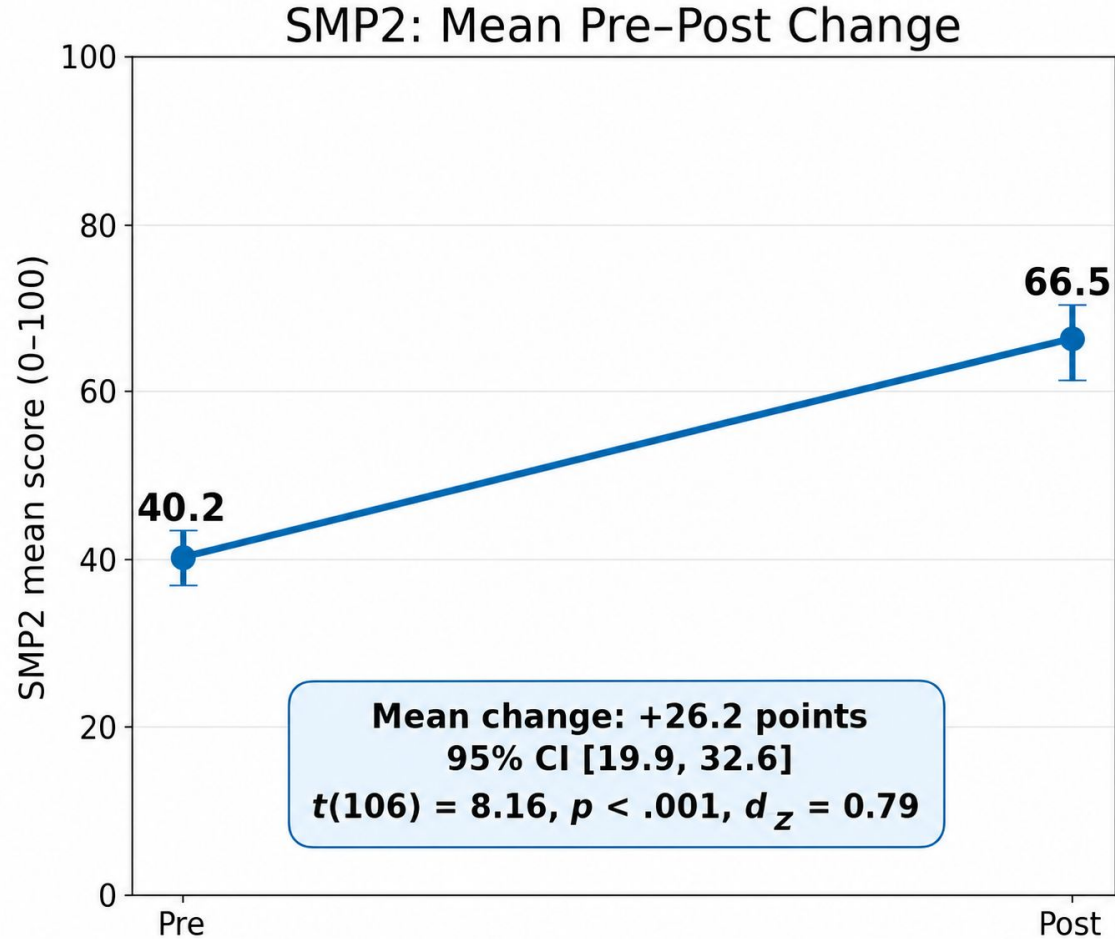
A paired-samples *t*-test compared student scores at pre and post.

Each student's Post score was compared with their own Pre score for SMP2: Reason abstractly and quantitatively and SMP8: Look for and express regularity in repeated reasoning.

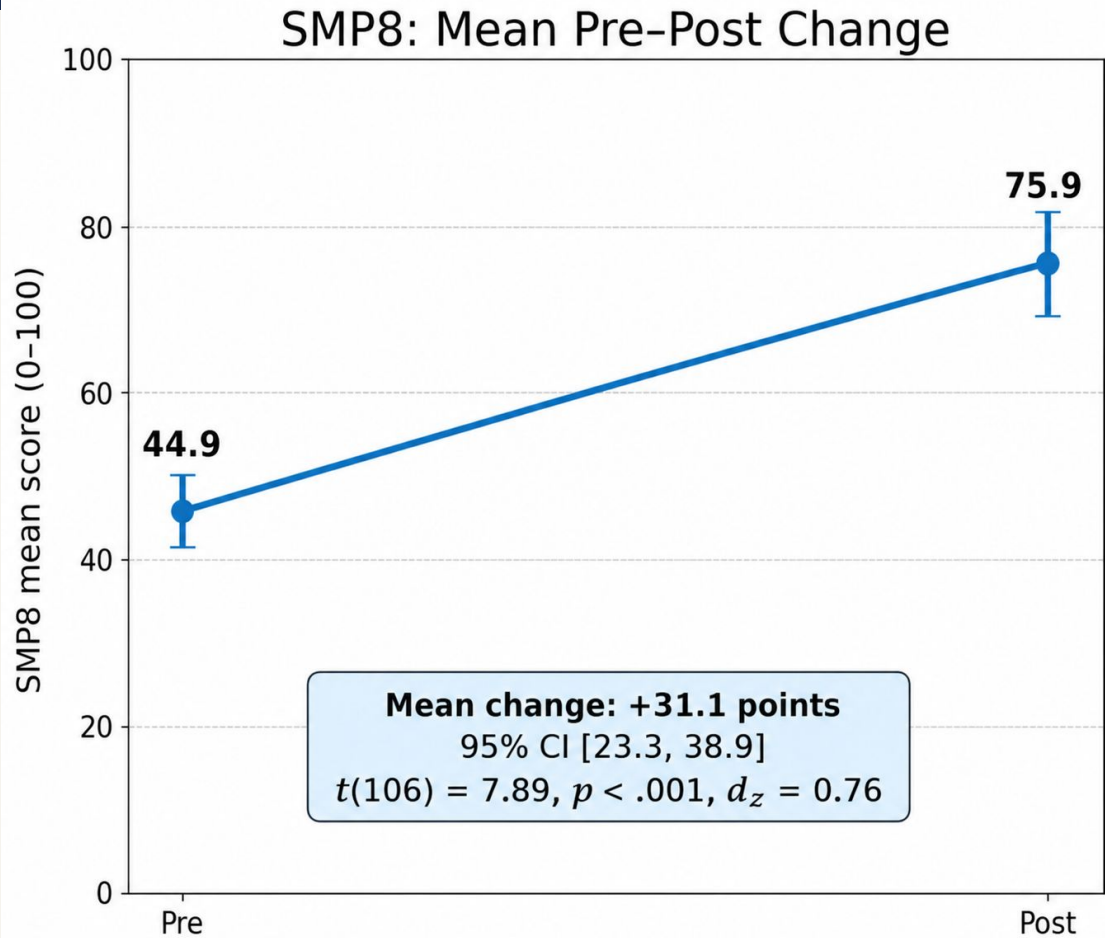
## **Follow-Up: Cross-Lagged Panel Model (CLPM, in the family of Structural Equation Modeling)**

CLPM asks whether earlier performance in one area predicts later performance in another area, while also accounting for students' earlier standing in that outcome. For example, does abstract & quantitative reasoning help predict later repeated reasoning beyond where students already started?

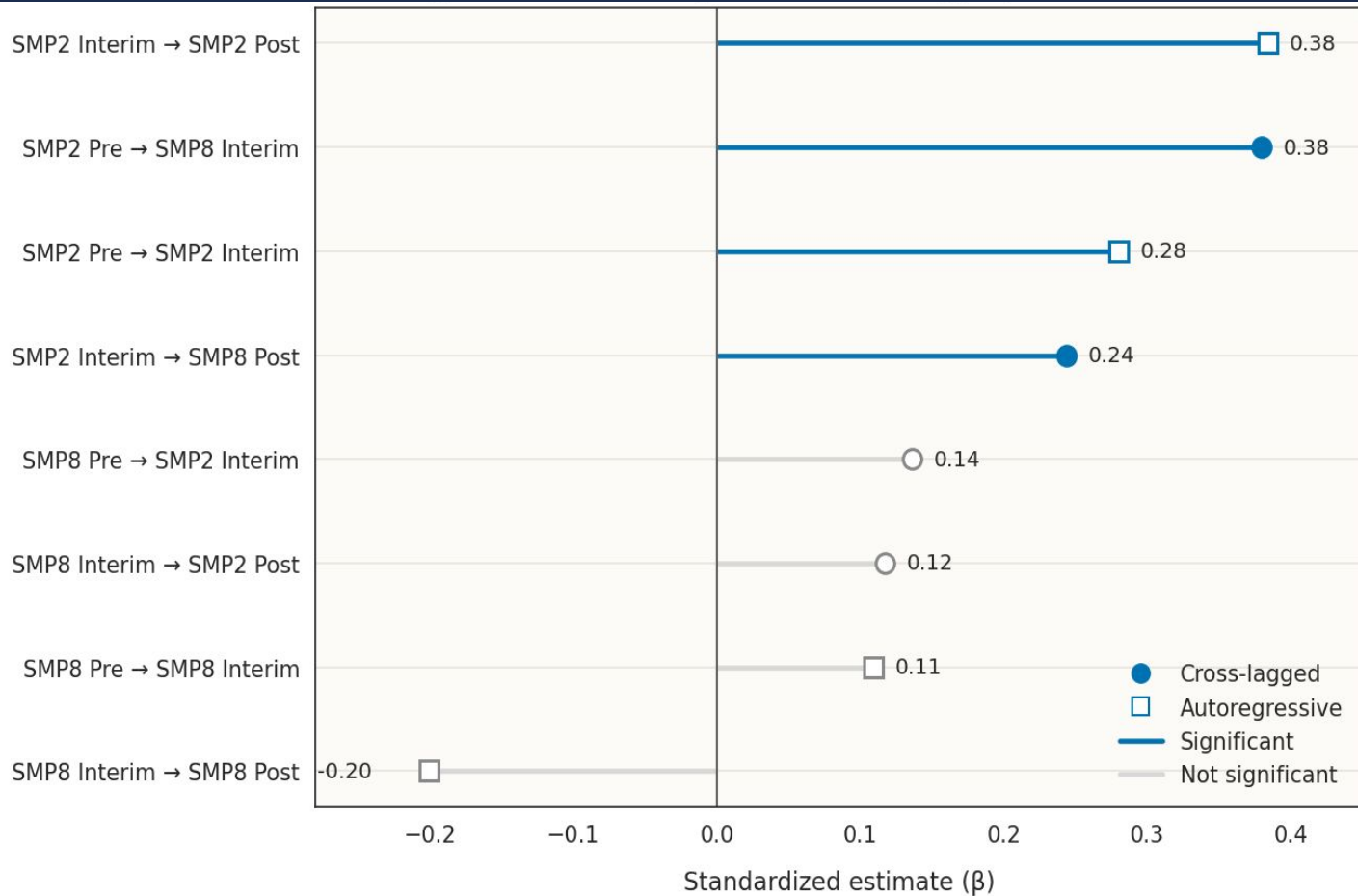
# Abstract & Quantitative Reasoning (SMP2)



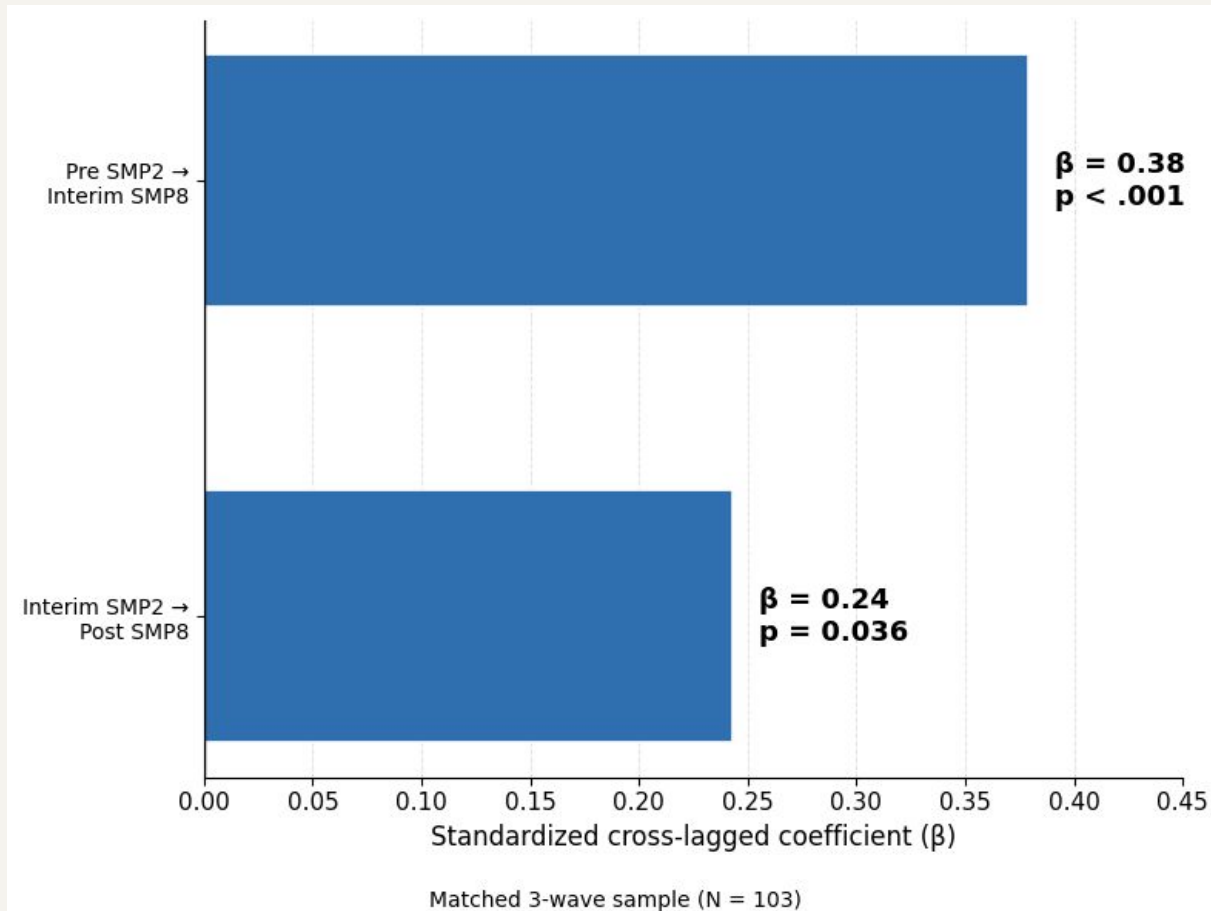
# Repeated Reasoning (SMP8)



# Abstract Reasoning Predicts Repeated Reasoning

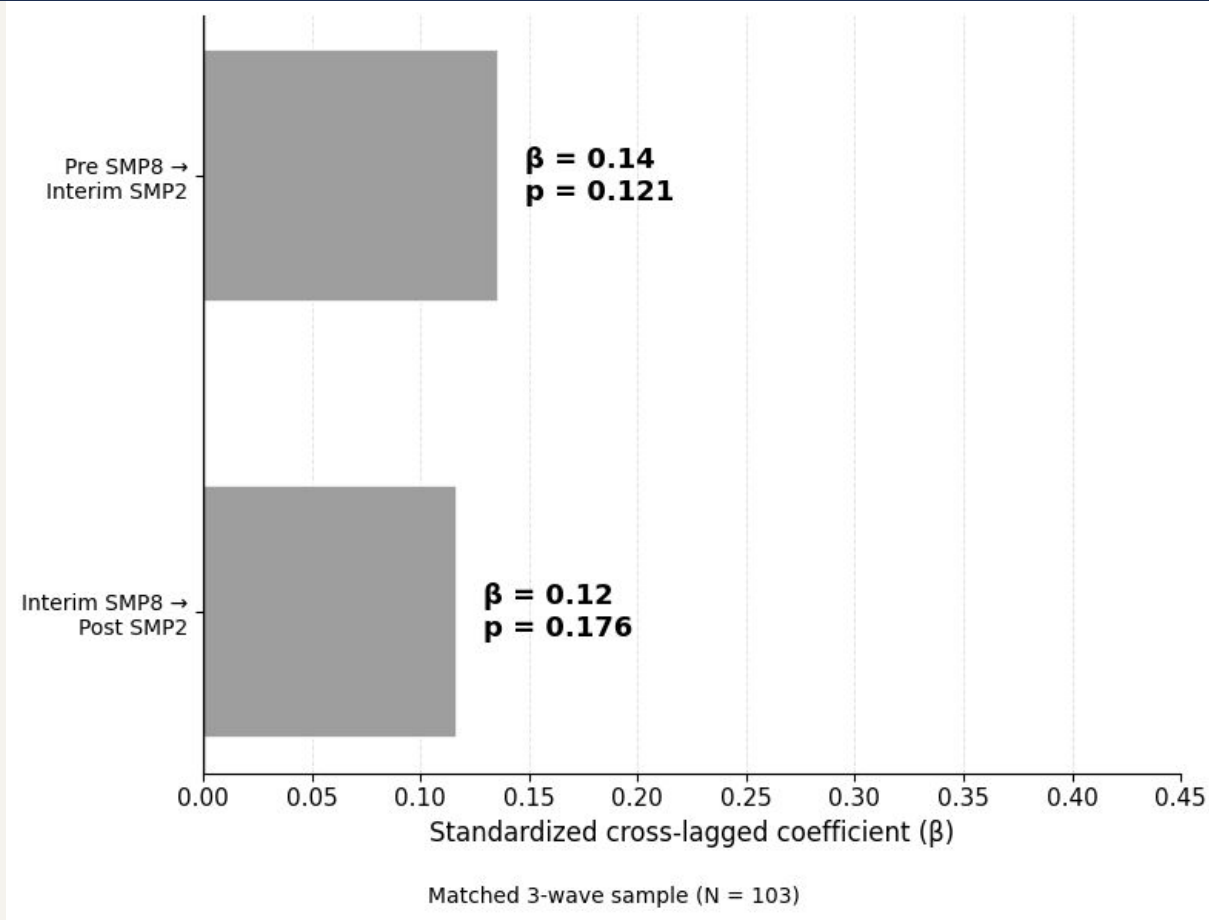


# Abstract Reasoning Predicts Repeated Reasoning





# Abstract Reasoning Predicts Repeated Reasoning



# Research Question 2: Literacy

How did mean literacy scores change during the program?

Did students differ in their growth trajectories?

## Latent Growth Curve Model

Estimates students' starting points and growth trajectories in literacy.

# Specific Literacy Standards

## California Department of Education

**W6.E.3a Write narratives to develop real or imagined experiences or events using effective technique, relevant descriptive details, and well-structured event sequences.**

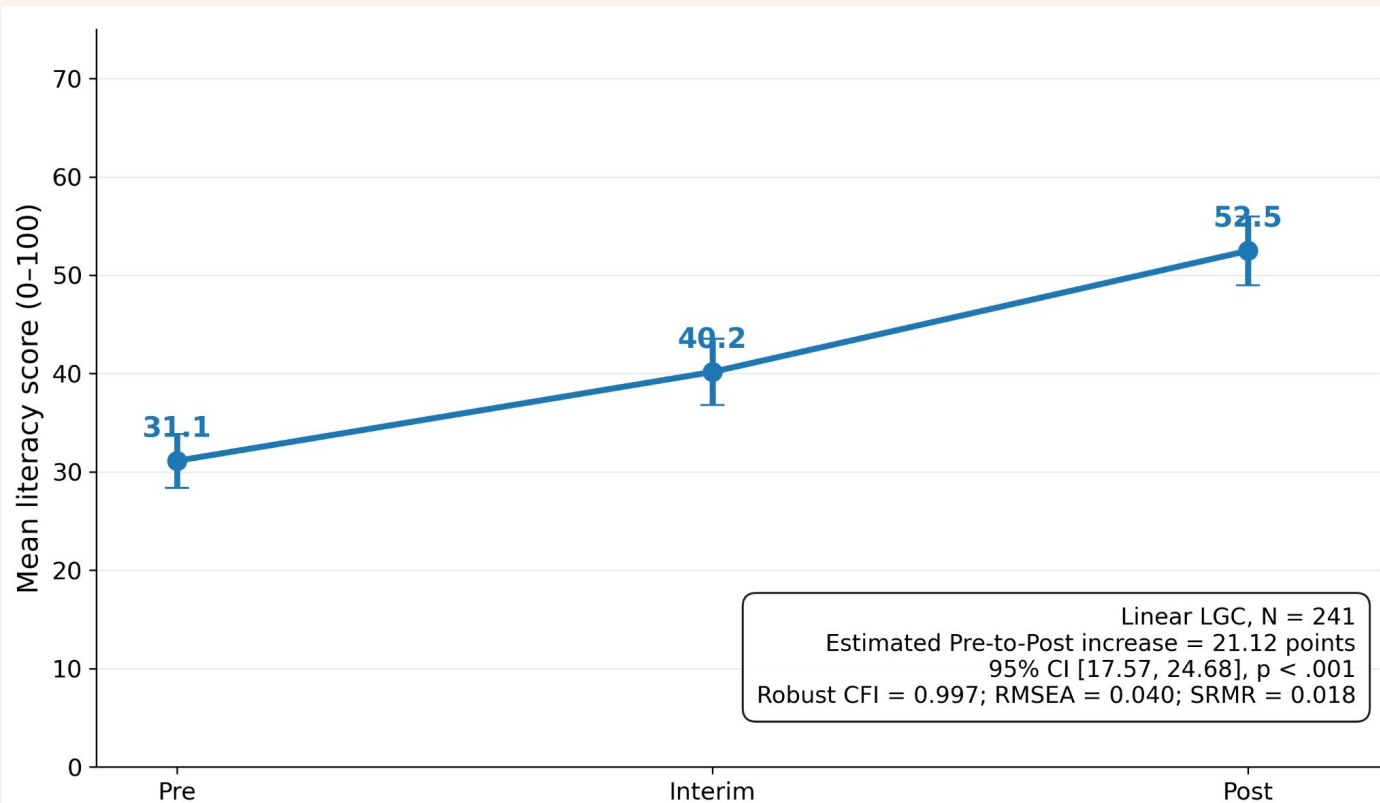
**W6.G.8 Research to Build and Present Knowledge**

**W6.A.6a Write informative/explanatory texts to examine a topic and convey ideas, concepts, and information through the selection, organization, and analysis of relevant content.**

# Overall Literacy Score

**Literacy scores represent each student's mean achievement percentage across the available writing-based literacy assessments at each wave. Individual literacy standards were not modeled separately.**

# Average Literacy Increased ( $p < .001$ )



- Estimated average Pre-to-Post growth: 21.12 points, 95% CI [17.57, 24.68],
- Not all growth identical: Ceiling effect (percentage scored test)

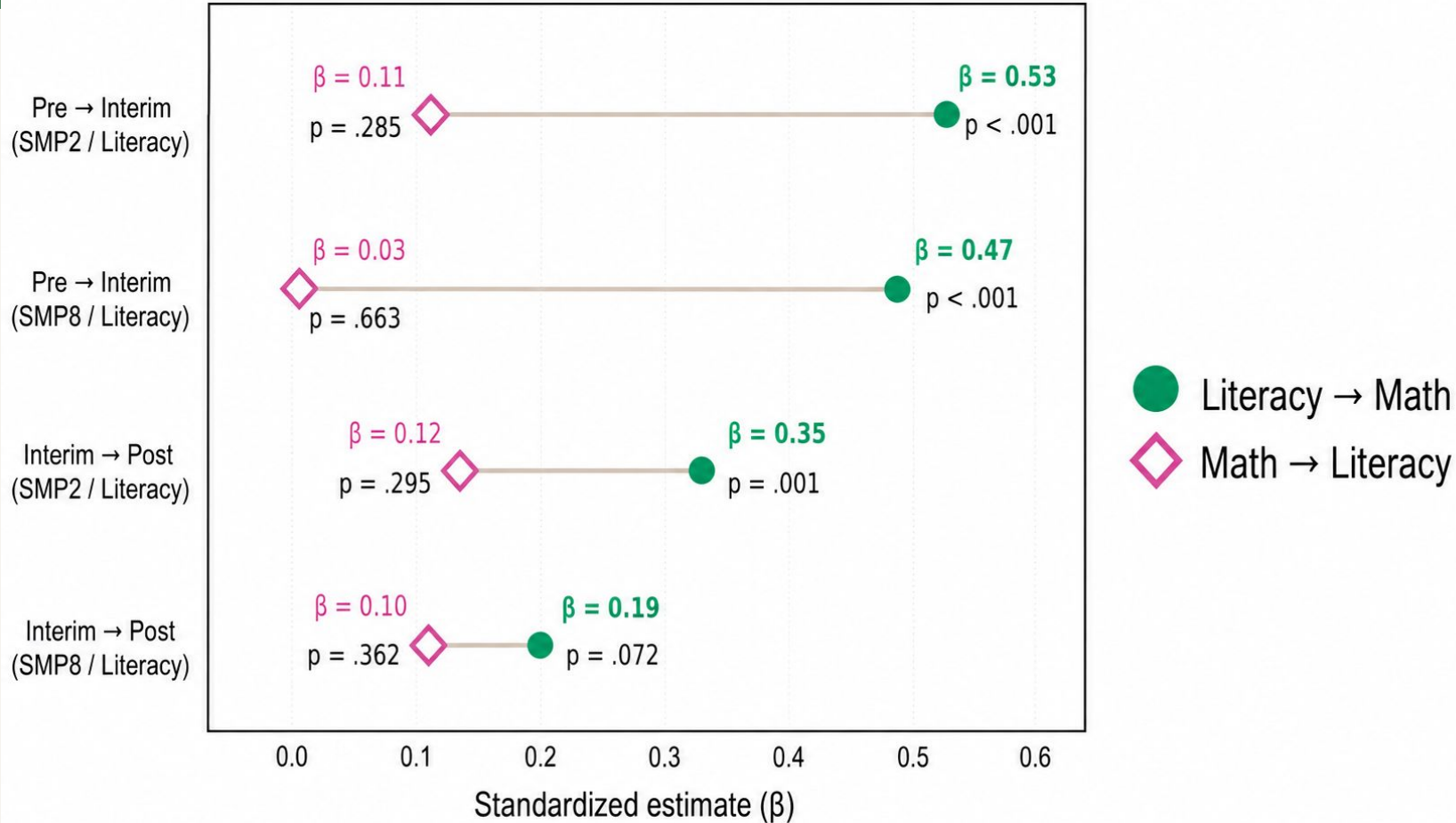
# Research Question 3: Math & Literacy

Do literacy scores predict later math growth?  
Do math scores predict later literacy growth?

## **Tri-Domain CLPM (Cross-Lagged Panel Model)**

- Does earlier literacy predict later math growth?
- Does earlier math predict later literacy growth?
- Controlling for students' prior scores in each domain.

# Earlier Literacy Scores Predicted Later Math Growth



# Research Question 4

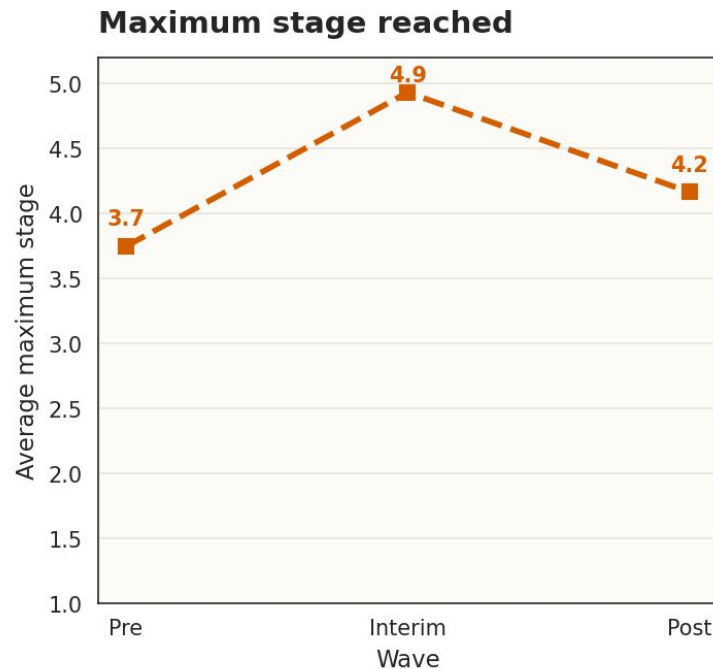
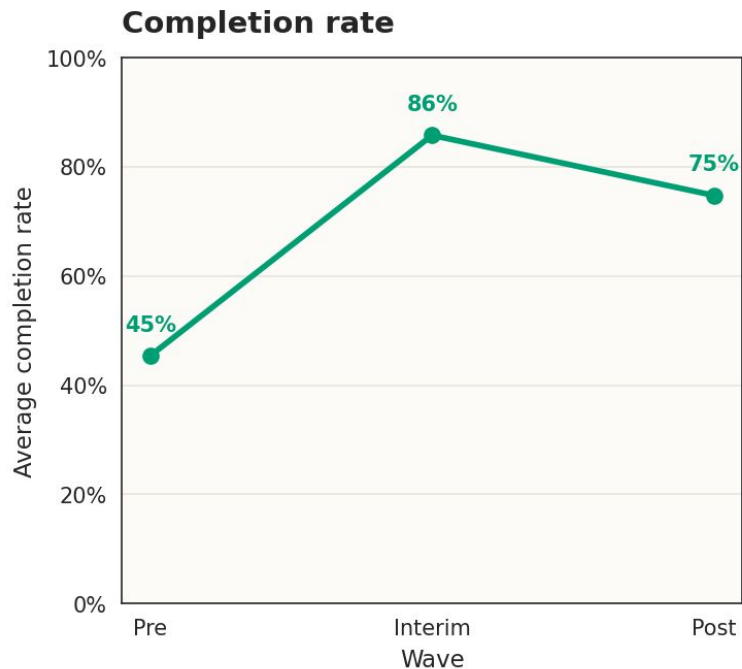
Do completion and stage attainment mediate the relationship between growth in academic domains?

## **CLPM with Mediator (Completion & Stage Attainment)**

Used to test whether completion & stage attainment helped explain the relationship between growth in math standards and literacy. This model did not fit well, so completion & stage attainment are better interpreted as part of the learning context rather than as a confirmed mechanism.



# Completion Rate & Stage Attainment



Engagement- mediation models fit the data poorly. Therefore, they did not provide reliable evidence that engagement mediated the longitudinal relationships between the standards. (RMSEA 0.18 much higher than ideal 0.05. CFI  $\approx$  0.84–0.86 lower than 0.9.)

Completion & stage attainment is better interpreted as part of the learning context than as a structural mechanism.

# Major Takeaway

- Literacy skills included within mathematics appeared to support mathematical achievement.

# THANK YOU for guidance and support!

- Boston College: Dr. Emma Klugman, Dr. Amy Semerjian, Cohort Peers
- Mindsets: Marissa di Pasquale, Dr. Aubrey Escobar
- Family

# Appendices

# Recommendations for Future Research

**1** Randomized control trial

**2** Tease out how literacy growth occurs within mathematics instruction

**3** Larger sample replication

**4** Additional measurement/instrumentation

# Useful takeaways from Mindsets!® program data

**1**

Earlier abstract and quantitative math reasoning was a predictor of later repeated math reasoning rather than the reverse.

**2**

Students grew in literacy with significant individual variation in growth.

**3**

Completion & stage attainment was high, especially at Interim.

**4**

Earlier literacy strongly predicted later math growth, whereas earlier mathematics did not significantly predict later literacy. This suggests that embedded literacy support may be especially important for students entering with weaker academic literacy.

# Academic Sources Which Support These Math and Literacy Connections

Erbeli, F., Shi, Q., Campbell, A. R., Hart, S. A., & Woltering, S. (2021). *Developmental dynamics between reading and math in elementary school*. *Developmental Science*, 24(1), e13004. <https://doi.org/10.1111/desc.13004>

These researchers found stronger reading performance was associated with later gains in math growth, particularly among children whose math performance was initially below average. They did not find the same pattern operating in the opposite direction.

Björn, P. M., Aunola, K., & Nurmi, J.-E. (2016). *Primary school text comprehension predicts mathematical word problem-solving skills in secondary school*. *Educational Psychology*, 36(2), 362–377. <https://doi.org/10.1080/01443410.2014.992392>

These researchers found that fourth-grade text comprehension predicted later mathematical word-problem performance even after the researchers controlled for reading fluency and basic calculation ability.

Akin, A. (2022). *Is reading comprehension associated with mathematics skills: A meta-analysis research*. *International Online Journal of Primary Education*, 11(1).

These researchers found a statistically significant, strong association between reading comprehension and mathematics skills. The relationship was strongest for mathematical problem solving and was particularly important for elementary students, students with learning disabilities, and second-language learners.

More studies show similar results.

# Support for Chosen Statistical Methods

## SEM

### Primary Models were SEM (Structural Equation Models)

Often used with math and literacy education data.

E.g. Asare, B. (2025). *Epistemological belief, reflective thinking and mathematics interest as a predictor of students' mathematics performance: mediation analysis via structural equation modeling (SEM)*. Cogent Education, 12(1). <https://doi.org/10.1080/2331186X.2025.2542437>

## Tri-D CLPM

### Tri-Domain CLPM (a type of SEM)

Examining whether earlier scores predict later scores after accounting for prior standing

## CLPM

### CLPM is used for longitudinal data

Controls for stability paths while checking how variables affect each other (e.g. SMP2 pre-SMP8 interim)

## MLR

### Robust Maximum Likelihood

MLR provides robust standard errors and scaled test statistics that reduce sensitivity to non-normality. It does not correct small samples, ceiling effects, range restriction, or model misspecification.



# Why use wave-specific z-scores in the CLPMs?

**Wave-specific z-scores were used for the CLPMs so standardized coefficients could be compared across domains.**

**Raw 0–100 scores were used for the latent growth models because wave-specific standardization removes mean growth over time.**

# Statistical Methods — Language Guide

## SEM

### Structural Equation Modeling (SEM)

The primary models were types of SEM, which let me test a whole system of relationships at once: how literacy, abstract math reasoning SMP2, repeated reasoning SMP8, and engagement relate to each other across time.

## LGC

### Latent Growth Curve Model

Estimates students' starting points and growth trajectories in literacy.

## CLPM

### Cross-Lagged Panel Model (a type of SEM)

CLPM asks whether earlier performance in one area predicts later performance in another area, while also accounting for students' earlier standing in that outcome. In other words, does abstract reasoning help predict later repeated reasoning beyond where students already started?

## Med

### CLPM with Engagement Mediator

Used to test whether engagement helped explain the relationship between growth in math standards. This model did not fit well, so engagement is better interpreted as part of the learning context rather than as a confirmed mechanism.

# Extra Research Questions

## **Parallel Process Latent Growth Curve Model**

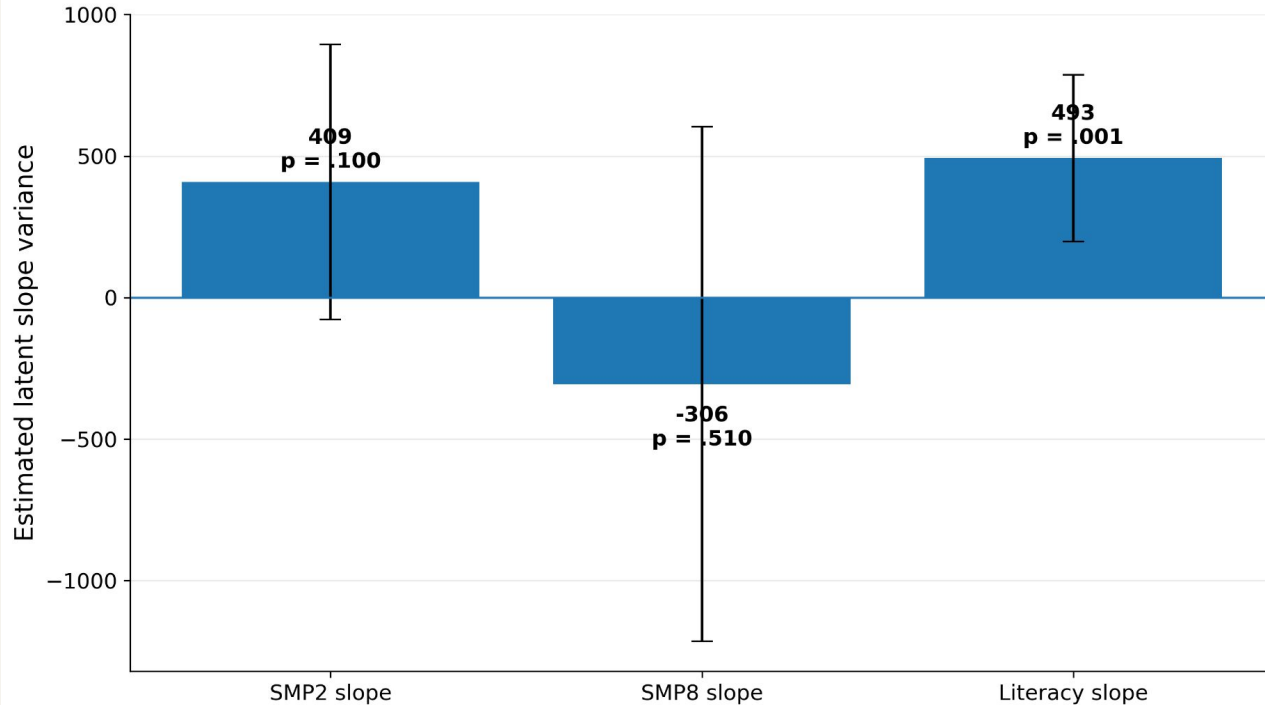
Do abstract reasoning, repeated reasoning, and literacy exhibit correlated developmental trajectories over the program?

## **Replaced with Exploratory Analysis of Spearman Correlations**

Exploratory analysis used because the latent growth solution was inadmissible

# The Joint Parallel-Process LGC Was Inadmissible

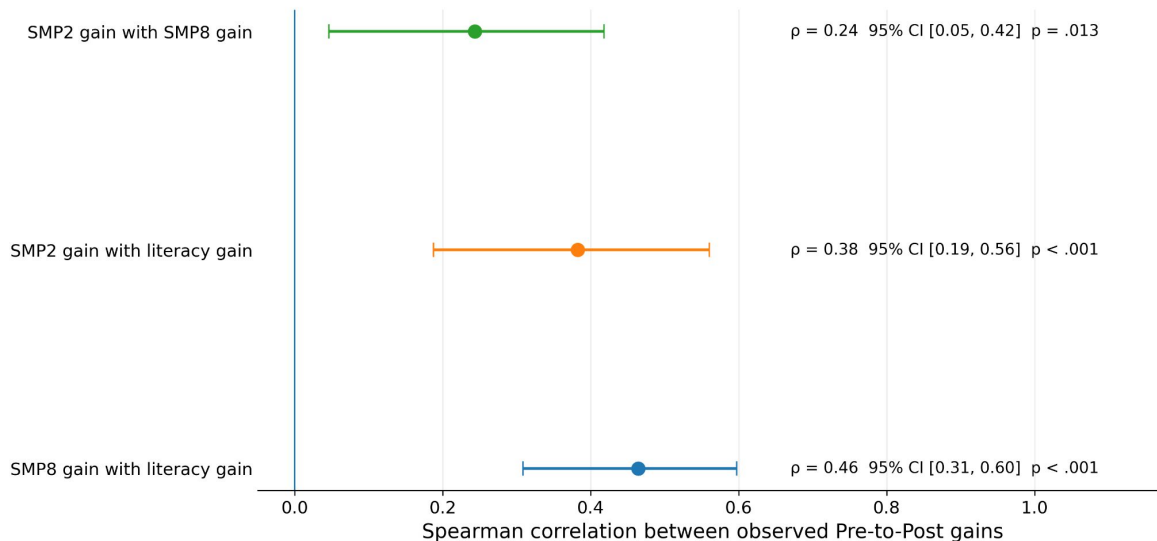
The SMP8 slope variance was negative, invalidating the joint LGC



Although the model converged, it failed admissibility checks; therefore, its latent slope correlations were not interpreted.

# Exploratory: Larger Observed Gains Tended to Co-Occur

## Larger observed gains tended to co-occur across domains



Exploratory sensitivity analysis, matched N = 103. Positive gain correlations do not establish correlated latent trajectories.

Spearman correlations among Pre-to-Post gain scores.

Positive observed-gain relationships provide exploratory evidence of co-occurring improvement.

Dots = strength of the relationship  
Lines = uncertainty around that estimate  
All lines stay above zero = all three relationships were positive and statistically significant